

KARSAKBAY, KazakhstanWMO: **356620**Lat: **47.833N**Lon: **66.717E**Elev: **488**StdP: **95.60**Time Zone: **6.00 (E06)**Period: **82-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.3	-24.9	-31.7	0.2	-26.4	-29.3	0.3	-23.8	17.9	-7.6	14.6	-7.8	1.0	270	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	33.9	17.4	32.1	16.9	30.3	16.6	19.7	30.2	18.4	29.2	17.4	27.9	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	11.8	24.9	14.2	10.7	23.6	13.0	9.9	22.7	58.5	30.1	54.1	29.0	50.9	27.7	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.2	7.8	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-13.0	-14.3	-7.5	6.9	14.1	20.5	22.9	20.5	13.4	4.9	-5.1	-10.4		
	DBStd	14.30	6.06	5.85	5.82	5.62	4.45	3.93	3.22	3.55	4.61	4.95	6.11	6.39		
	HDD10.0	3349	713	681	543	121	16	0	0	0	18	170	454	633		
	HDD18.3	5387	971	914	802	343	142	22	5	19	158	415	704	892		
	CDD10.0	1341	0	0	0	29	142	314	398	324	120	13	0	0		
	CDD18.3	336	0	0	0	0	9	87	145	86	9	0	0	0		
	CDH23.3	3981	0	0	0	8	127	1076	1664	971	132	3	0	0		
	CDH26.7	1535	0	0	0	1	16	417	714	359	28	0	0	0		
(14) Wind	WSAvg	2.4	2.4	2.6	2.7	2.8	2.7	2.4	2.4	2.1	2.0	2.2	2.4	2.0		
(15) Precipitation	PrecAvg	190	22	11	15	20	14	15	19	9	13	21	16	18		
	PrecMax	337	91	30	37	86	36	53	83	33	48	63	56	52		
	PrecMin	108	2	0	0	0	0	0	1	0	0	0	1	1		
	PrecStd	56	21	8	10	21	11	13	20	9	14	16	12	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	-0.5	9.4	24.4	28.8	36.5	36.4	35.6	30.1	21.5	9.2	4.4		
		MCWB	-0.5	-1.4	6.3	12.9	15.6	18.2	17.8	17.7	16.5	11.2	5.9	1.2		
	2%	DB	-1.2	-2.1	4.5	20.8	26.4	33.1	34.3	32.6	26.8	18.4	6.0	0.9		
		MCWB	-1.7	-3.0	2.7	11.5	14.3	17.5	17.3	16.7	14.2	9.2	3.1	0.2		
	5%	DB	-2.9	-3.5	1.8	18.4	24.7	31.0	32.5	30.6	24.5	15.9	4.0	-0.6		
		MCWB	-3.5	-4.4	0.4	9.9	14.0	17.0	17.2	16.0	13.0	8.5	2.0	-1.5		
	10%	DB	-4.2	-5.2	0.0	16.1	22.9	29.0	30.7	28.5	22.3	13.0	2.1	-2.1		
		MCWB	-4.8	-6.1	-1.1	8.7	13.0	16.8	16.9	15.3	11.8	6.9	0.7	-3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	-1.3	6.2	14.2	17.5	27.3	21.1	20.7	17.9	12.2	6.8	1.5		
		MCDB	0.2	-1.0	8.5	22.1	25.5	28.4	31.8	31.9	27.1	18.3	9.1	3.5		
	2%	WB	-1.6	-3.0	2.8	12.1	15.7	20.0	19.6	17.9	15.2	10.1	3.8	0.2		
		MCDB	-1.3	-2.2	4.4	18.9	23.9	30.1	30.3	29.6	24.7	16.9	5.4	1.2		
	5%	WB	-3.4	-4.3	0.6	10.6	14.5	18.4	18.6	16.7	13.6	8.9	2.4	-1.4		
		MCDB	-2.9	-3.5	1.7	17.3	23.1	28.9	29.4	28.0	23.5	14.7	3.8	-0.6		
	10%	WB	-4.9	-6.0	-1.0	9.3	13.4	17.2	17.6	15.9	12.3	7.5	0.8	-3.0		
		MCDB	-4.3	-5.2	0.0	15.2	21.7	27.1	28.3	26.9	21.6	12.5	2.3	-2.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	8.0	8.1	10.9	12.1	13.0	12.4	12.6	12.4	10.6	6.9	6.9		
		MCDBR	6.8	8.4	8.5	14.9	14.8	15.5	15.3	15.3	15.6	15.1	8.9	7.9		
	5% WB	MCWBR	6.2	7.3	6.8	7.6	7.2	7.1	6.1	6.5	8.0	8.1	7.0	6.9		
		MCWBR	6.3	7.9	8.1	13.7	13.3	13.7	13.1	13.3	14.9	13.1	7.7	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.329	0.387	0.420	0.428	0.419	0.418	0.393	0.350	0.347	0.329	0.302		
	taud		2.100	2.010	2.046	2.079	2.117	2.209	2.234	2.284	2.359	2.281	2.216	2.145		
	Ebn at Noon		746	800	813	828	836	846	841	847	854	785	706	686		
	Edh at Noon		89	123	140	150	150	138	133	121	102	93	80	76		
(44) All-Sky Solar Radiation	RadAvg		1.54	2.67	3.85	5.21	6.60	6.99	6.66	6.07	4.71	2.78	1.44	1.11		
	RadStd		0.25	0.28	0.38	0.48	0.45	0.41	0.33	0.35	0.17	0.25	0.14	0.19		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page